

## Experiment No. 1

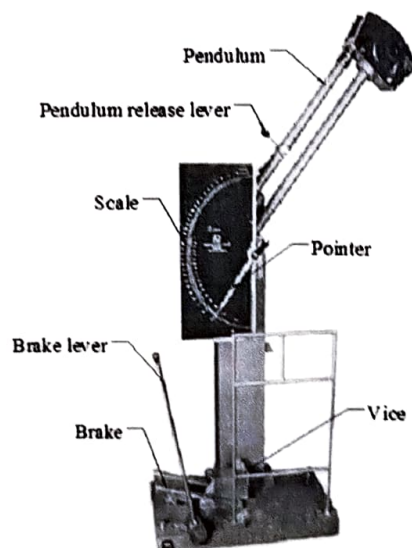
**Title:** Determination of impact strength by Charpy impact test

**Objective:** To determine the resistance of shock or impact strength of a given specimen by Charpy impact test

**Equipments used:** Impact testing machine

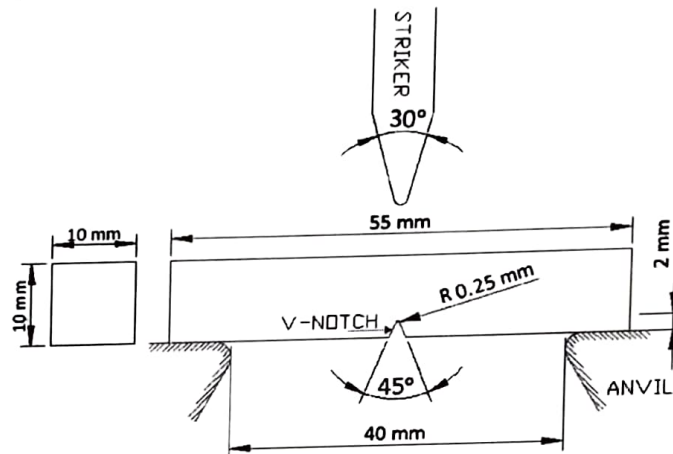
### **Theory:**

The impact is a strong force or shock which occurs over a short time period when two or more bodies collide. Such a force or acceleration usually has a greater effect than a weaker force applied over a proportionally longer period. The impact test is performed to find out the energy needed to fracture a material and its toughness. The toughness of a material is its ability to absorb energy in plastic range especially when subjected to dynamic loading. The impact test can also be used to determine the ductility of a material. If the fracture surface is a flat plane, the material is brittle, and if the material breaks with jagged edges or shear lips, then the material is ductile. Usually a material does not break in just one way or the other, and thus comparing the jagged to flat surface areas of the fracture will give an estimate of the percentage of ductile and brittle fracture.



**Fig. 1** Impact testing machine

In Charpy impact test, the specimen is placed as 'simply supported beam' between two supports with a notch equidistant between them. It is struck by the pendulum released from an angle of  $135^\circ$ , on the side opposite to the notch. In this test two types of specimen are in common use - the keyhole notch specimen and the V notch specimen. In this experiment the V-notch specimen is used and it is a bar of  $10\text{ mm} \times 10\text{ mm}$  cross section which has a  $45^\circ$  V-notch, 2 mm deep with a 0.25 mm root radius as shown in Fig 2. The specimen is supported as a beam in horizontal position, with span 40 mm, and is struck behind the notch by the impact of a heavy swinging pendulum.



**Fig. 2** Dimensions of specimen for Charpy impact test

The Notch Impact strength ( $I$ ) is calculated as per the following formula

$$I = \frac{K}{A},$$

where,  $I$  = Impact strength in  $\text{J/m}^2$

$K$  = Impact energy absorbed by the specimen in  $\text{J}$

$A$  = Area of cross section of specimen below the notch before test in  $\text{m}^2$

**Procedure:**

1. The scale of the machine is calibrated by releasing the hammer from the clamp without placing any specimen on the anvil.
2. The hammer is brought back to its initial energy level and clamped.
3. The specimen is fixed as a simply supported beam in the anvil.
4. The scale is adjusted to read zero.

5. The hammer is released from the clamp.
6. The energy absorbed by the specimen during fracture is noted from the scale.

**Observation and Calculation:**

Specimen Material = \_\_\_\_\_  
Impact energy absorbed (reading on the scale) 'K' = \_\_\_\_\_ J  
Area of cross section of specimen below the notch 'A' = \_\_\_\_\_ m<sup>2</sup>  
The notch impact strength 'I' = \_\_\_\_\_ J/m<sup>2</sup>

**Conclusion:**

From this experiment we found,

Impact Strength (I) for the specimen by Charpy impact test is \_\_\_\_\_ J/m<sup>2</sup>

### Experiment No. 3

**Title:** Determination of hardness by Brinell hardness test

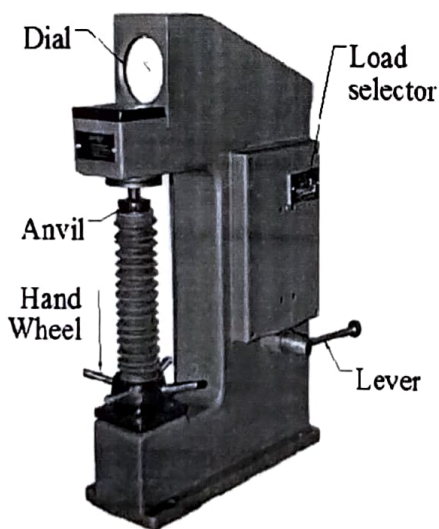
**Objective:** To determine the hardness of a given specimen by Brinell hardness test

**Equipments used:** Hardness Testing Machine

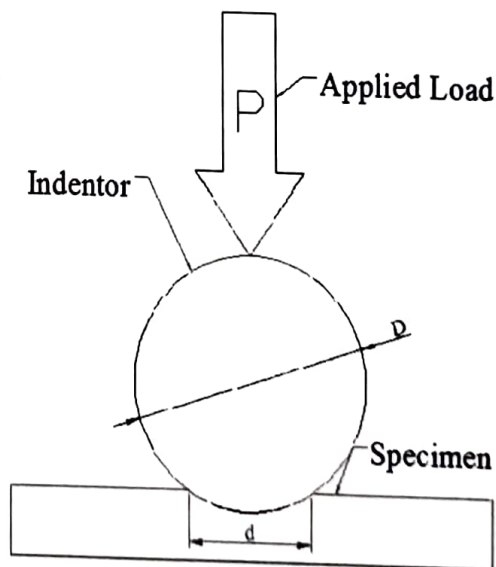
#### **Theory:**

Hardness is defined as the resistance against plastic deformation or ability to resist abrasion, scratching and indentation. In Brinell hardness test a hard steel ball of diameter ' $D$ ' is pressed into the surface of the specimen by a load ' $P$ ', which gives an indentation mark of the steel ball on the surface as indicated on Fig 2. The diameter of the indentation ' $d$ ' is measured with the help of a microscope and the Brinell hardness number (BHN) is found out by the following equation,

$$\begin{aligned} \text{BHN} &= \frac{\text{Applied load}}{\text{Area of Indentation}} \\ &= \frac{2P}{\pi D(D - \sqrt{D^2 - d^2})} \end{aligned}$$



**Fig. 1** Hardness testing machine



**Fig. 2** Brinell hardness test

**Procedure:**

1. Clean the surface of the material so that it is free from dirt.
2. Put the steel ball indenter (of suitable diameter) into position.
3. Keep the lever at position 'A'.
4. Place specimen securely on anvil and set the load.
5. Turn the hand wheel clockwise, so that specimen will push the indenter and show a reading on dial gauge, as small pointer reaches at '3' (red spot) and long pointer reaches at '0' or 'Set' position.
6. Turn the lever from position 'A' to 'B' slowly so that the total load is brought into action without any jerks. The long pointer of dial gauge reaches a steady position when indentation is complete.
7. The load is allowed to act for 10 – 15 seconds.
8. Take back the lever to position 'A' slowly.
9. The specimen is taken out by turning the hand wheel anticlockwise.
10. The diameter of the impression is measured using microscope.

**Observation Table:**

| Sl. No. | Diameter of steel ball 'D' (mm) | Load applied 'P' (kgf) | Diameter of indentation 'd' (mm) | BHN |
|---------|---------------------------------|------------------------|----------------------------------|-----|
|         |                                 |                        |                                  |     |

**Calculation:**

$$\begin{aligned} \text{BHN} &= \frac{\text{Load applied (P)}}{\text{Spherical area of impression}} \\ &= \frac{2P}{\pi D (D - \sqrt{D^2 - d^2})} \end{aligned}$$

**Conclusion:**

From the experiment we found,

The Brinell hardness number (BHN) of the specimen is \_\_\_\_\_ kgf/mm<sup>2</sup>

## Experiment No. 2

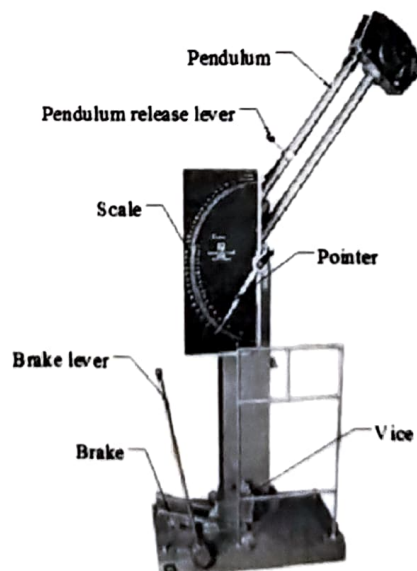
**Title:** Determination of impact strength by Izod Impact Test

**Objective:** To determine the resistance of shock or impact strength of a given specimen by Izod impact test

**Equipments used:** Impact testing machine

### **Theory:**

The impact is a strong force or shock which occurs over a short time period when two or more bodies collide. Such a force or acceleration usually has a greater effect than a weaker force applied over a proportionally longer period. The impact test is performed to find out the energy needed to fracture a material and its *toughness*. The *toughness* of a material is its ability to absorb energy in plastic range especially when subjected to dynamic loading. The impact test can also be used to determine the ductility of a material. If the fracture surface is a flat plane, the material is brittle, and if the material breaks with jagged edges or shear lips, then the material is ductile. Usually a material does not break in just one way or the other, and thus comparing the jagged to flat surface areas of the fracture will give an estimate of the percentage of ductile and brittle fracture.



**Fig 1 Impact testing machine**

In Izod impact test, the specimen is held in a vice as a cantilever beam, with the center line of the notch at the same level as the top surface of the clamping device. It is struck by the pendulum released from an angle of  $90^\circ$  on the side of the notch 22 mm above the edge of the clamp. In this test two types of specimen are in common use - the keyhole notch specimen and the V notch specimen. In this experiment the V-notch specimen is used and it is a bar of  $10\text{ mm} \times 10\text{ mm}$  cross section which has a  $45^\circ$  V-notch, 2 mm deep with a 0.25 mm root radius as shown in Fig 2.

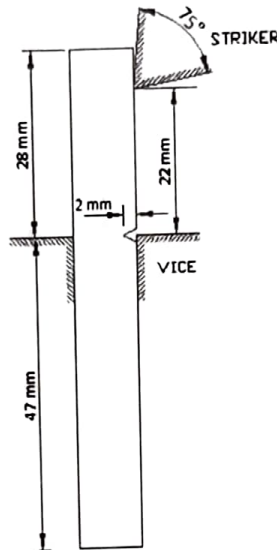


Fig. 2 Dimensions of specimen for Izod impact test

The Notch Impact strength (I) is calculated as per the following formula

$$I = \frac{K}{A}$$

where,

I = Impact strength in  $\text{J/m}^2$

K = Impact energy absorbed by the specimen in J

A = Area of cross section of specimen below the notch in  $\text{m}^2$

#### Procedure:

1. The scale of the machine is calibrated by releasing the hammer from the clamp without placing any specimen on the anvil.
2. The hammer is brought back to its initial energy level and clamped.
3. The specimen is fixed as a simply supported beam in the anvil.

4. The scale is adjusted to read zero.
5. The hammer is released from the clamp.
6. The energy absorbed by the specimen during fracture is noted from the scale.

**Observation and calculation:**

|                                                       |                          |
|-------------------------------------------------------|--------------------------|
| Specimen material                                     | = _____                  |
| Impact energy absorbed (reading on the scale) 'K'     | = _____ J                |
| Area of cross section of specimen below the notch 'A' | = _____ m <sup>2</sup>   |
| The notch impact strength 'I'                         | = _____ J/m <sup>2</sup> |

**Conclusion:**

From this experiment we found,

Impact Strength (I) for the specimen by Izod test is \_\_\_\_\_ J/m<sup>2</sup>

### **Experiment No. 5**

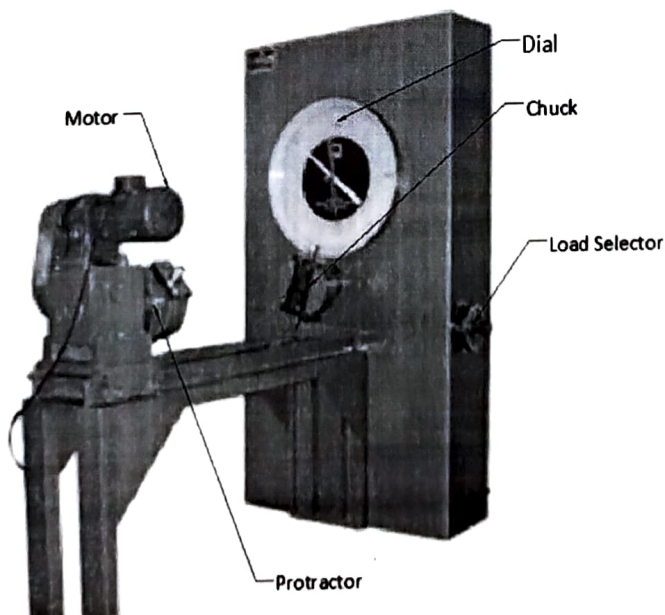
**Title:** Determination of modulus of rigidity, torsional stiffness and ultimate shear strength by Torsion Test

**Objective:** To determine the Modulus of Rigidity, Torsional Stiffness and Ultimate Shear Strength of a given specimen

**Equipments Used:** Torsion Testing Machine

#### **Theory:**

In workshops and factories, a turning force is always applied to transmit energy by rotation. This turning force is applied to a suitable point at some distance from the axis of the shaft. The product of this turning force, and the distance between the point of application of the force and the axis of the shaft is known as torque, which is also known as turning moment or twisting moment. So whenever a twisting force or torque is applied on a shaft, then the shaft is said to be subjected to torsion. Due to this torque, every cross section of the shaft is subjected to some shear stress.



**Fig 1 Torsion testing machine**

In Torsion test, if a solid shaft of diameter 'D' is subjected to torque 'T' whose angle of twist is 'θ' (as shown in Fig. 2), then modulus of rigidity, shear stress and torsional stiffness of the shaft is calculated by the torsion equation as follows:

$$\frac{T}{J} = \frac{\tau}{R} = \frac{G\theta}{L}$$

where,

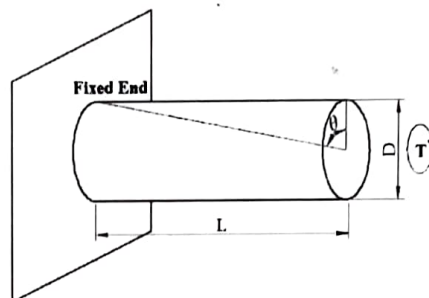
J = Polar moment of inertia of the cross-section of the shaft

$$= \frac{\pi D^4}{32} \text{ mm}^4$$

τ = Shear stress on the shaft

L = Length of the shaft (gauge length)

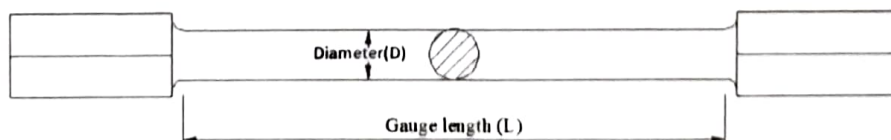
G = Modulus of rigidity of the shaft



**Fig. 2** Shaft under torsion

**Procedure:**

1. Measure the diameter and gauge length of the specimen as shown in Fig 3.
2. Fix the specimen in the jaws of the machine.
3. Set the pointer on the torque measuring scale to zero.
4. Switch on the machine and start the motor.
5. Note down the torque and angle of twist for every five degrees, till fracture.
6. Plot a graph of torque (T) vs. angle of twist (θ).
7. Now calculate modulus of rigidity, shear stress and torsional stiffness of the specimen.



**Fig 3** Specimen

**Observation Table No 1:**

| Sl. No. | Diameter 'D' (mm) |     |       |      | Gauge Length 'L' (mm) | Polar Moment of Inertia 'J' (mm <sup>4</sup> ) |
|---------|-------------------|-----|-------|------|-----------------------|------------------------------------------------|
|         | MSR               | VSR | Total | Mean |                       |                                                |
|         |                   |     |       |      |                       |                                                |

**Observation Table No 2:**

| Sl. No. | Angle of Twist 'θ' (rad) | Torque 'T' (N-mm) | Modulus of Rigidity<br>$G = \frac{T}{\theta} \times \frac{L}{J}$<br>(N/mm <sup>2</sup> ) | Shear stress<br>$\tau = \frac{G\theta R}{L}$<br>(N/mm <sup>2</sup> ) | Torsional Stiffness,<br>$K = \frac{T}{\theta} = \frac{GJ}{L}$<br>(N-mm/rad) |
|---------|--------------------------|-------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------|
|         |                          |                   |                                                                                          |                                                                      |                                                                             |

**Conclusion:**

From the test, we find that:

Modulus of Rigidity of the specimen = \_\_\_\_\_ N/mm<sup>2</sup>

Shear stress of the specimen = \_\_\_\_\_ N/mm<sup>2</sup>

Torsional Stiffness of the specimen = \_\_\_\_\_ N-mm/rad

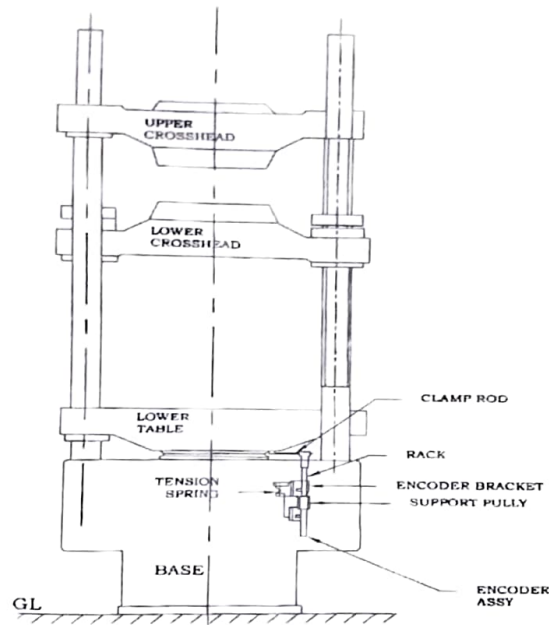
Ultimate Shear Strength of the specimen = \_\_\_\_\_ N/mm<sup>2</sup>

**Experiment No. 2**

**Title:** Determination of compressive strength of a concrete cube

**Objective:** To find out the compressive strength of a concrete cube by compression test

**Equipment used:** Universal testing machine (UTM)



**Fig. 1** Schematic diagram of Universal Testing Machine

**Theory:**

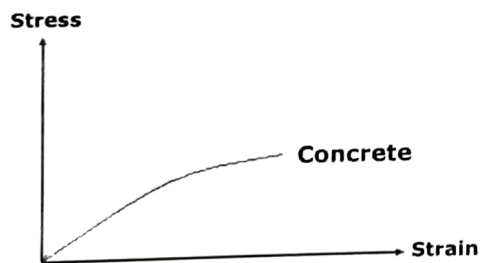
A compression test determines behavior of materials under crushing loads. Several machines and structure components such as columns and struts are subjected to compressive load. These components are made of high compressive strength materials. Not all the materials are strong in compression. Several materials, which are good in tension, are poor in compression. Contrary to this, many materials poor in tension but very strong in compression. That is why determination of ultimate compressive strength is essential before using a material. This strength is determined by conduct of a compression test which is just opposite in nature to tensile test. Nature of deformation and fracture is quite different from that in tensile test. In this test a compressive load is applied on the cube which tends to squeeze the specimen. The load is applied on the cube till its failure. The

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**GIRIJANANDA CHOWDHURY University**

ratio between the load at failure and the cross-sectional area of the cube will give the ultimate compressive strength of the cube.

**Procedure:**

1. Measure the dimension of the cube and calculate the cross-section area
2. Place the cube centrally on the lower table, such that the centre of the lower crosshead is vertically above the centre of specimen.
3. Close the release valve on the machine and turn on the machine
4. Now apply load on the specimen by turning slowly the load valve in anticlockwise direction until the specimen fails
5. Note down the load at which the specimen breaks from the display of the machine and open the release valve.
6. Now draw a stress-strain curve by using the values obtained during the test



**Fig. 2** Stress-Strain curve for concrete

**Observation & Calculation:**

- |                                                                             |                                   |
|-----------------------------------------------------------------------------|-----------------------------------|
| 1. Length of the cube 'l'                                                   | = _____ mm                        |
| 2. Breadth of the cube 'b'                                                  | = _____ mm                        |
| 3. Load taken by the cube at the time of failure 'W'                        | = _____ N                         |
| 4. Cross-section area of the cube 'A'                                       | = _____ mm <sup>2</sup>           |
| 5. Compressive strength or Ultimate strength of the cube ' $\sigma_{max}$ ' | = $\frac{W}{A}$ N/mm <sup>2</sup> |

**Result:** The compressive strength of given concrete cube is \_\_\_\_\_ N/mm<sup>2</sup>

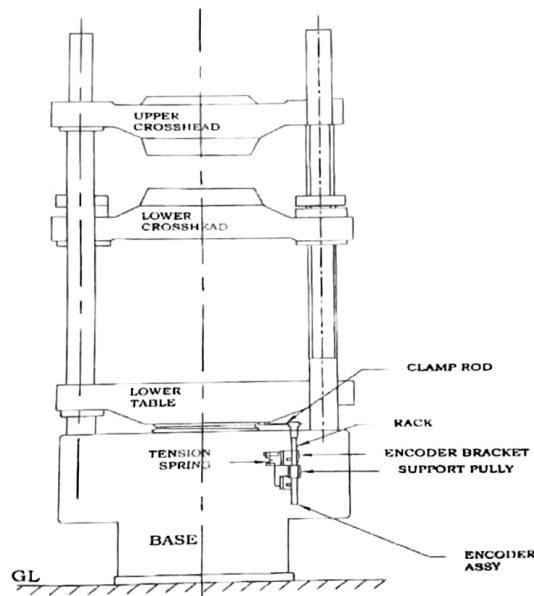
**Experiment No. 1**

**Title:** PERFORM TENSION TEST ON REINFORCING STEEL BARS.

**Object:** - To conduct a tensile test on a mild steel specimen and determine the following:

- i) Ultimate strength ii) Upper yield strength iii) Lower yield strength
- iv) Percentage elongation

**Equipment used:** Universal testing machine



**Fig. 1 Schematic of Universal Testing Machine**

**Description:**

For structural steel and reinforcing bars, tensile test, a destructive material test, is the most important. Tensile test provides relevant information for evaluation of the tensile properties and for quality assurance purposes. One usually conducts a tensile test on a standard specimen of the material to obtain the stress-strain diagram of a material. The diagram representing the relation between stress and strain in a given material is an important characteristic of the material. Important basic properties, viz., the proportional limit, elastic limit, the yield strength, the ultimate stress, the elongation at fracture, young's modulus, etc. can be obtained from the stress-strain diagram. An idealized stress strain curve for steel is shown in Figure 2. An axially loaded tension member exhibits an elastic range (a-b) ending at yielding (b), followed by yield plateau (b-c). In the yield plateau, the load almost remains constant as the elongation increases. On further loading, the material exhibits a smaller increase in tension due to reorientation of the crystal structure of the metal with elongation (c-d). This region is referred to as the strain hardening range. After reaching the ultimate load (d), the loading decreases as the elongation increases (d-e), until the specimen eventually fractures at (e). The region (d-e) is referred to as the

necking range. Since ductile materials are strong in tension, the failure is caused by shear stresses. The direction of the shear makes an angle of  $45^\circ$  with the applied stress. Thus, the plane of failure is at  $45^\circ$  from the axis of the shaft, and the failure is known as cup and cone failure because of the fracture surface.

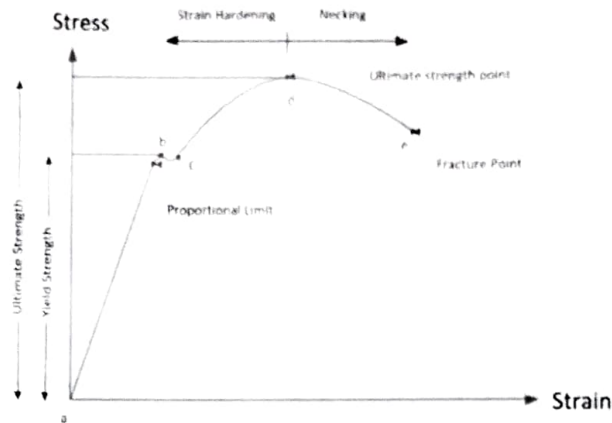


Figure 2: Engineering Stress-Strain Curve of metallic materials.

The following are the properties of steel determined from Stress-strain curve:

- **Proportional limit** — It is the region in the stress-strain curve which obeys the Hooke's Law. In this region, the ratio of stress with strain gives a constant called Young's Modulus.
- **Modulus of elasticity or Young's Modulus,  $E$**  — It is a measure of a material's axial stiffness; it does not change with the type of steel and has the value of  $2.5 \times 10^5$  MPa.
- **Elastic limit** — It is the point in the graph up to which the material returns to its original position when the load acting on it is completely removed.
- **Yield stress,  $\sigma_y$**  — It is the stress at which elongation first occurs in the test piece without increasing the load during the tensile test. In the case of steels with no such definite yield point, proof stress shall be applicable.
- **Ultimate or tensile strength,  $\sigma_u$**  — It is the maximum load reached in a tensile test divided by the initial cross-sectional area. It is the largest value of stress that the material can support. Ideally, the yield stress and ultimate strength should be calculated based on reduced area of cross section, but it is not practical to determine reduced area of cross section at various stages of loading. Thus, this stress is also called the engineering stress.
- **Fracture Stress** — It is the stress at which the failure of the material takes place.
- **Percentage permanent elongation (after fracture):** The amount of permanent strain from proportionality limit to the point of rupture or fracture is a measure of ductility. It is measured by computing the percentage elongation of the test specimen over the specified or standard gauge length. It is given by:  $\text{Percentage permanent elongation} = (L_u - L_0) / L_0 \times 100$   
where  $L_0$  is the gauge length before application of force and  $L_u$  is the gauge length after rupture of the test piece. The two pieces having been carefully fitted back together so that their axes lie in a straight line.
- **Percentage Total Elongation at Maximum Force** — It is the percentage elongation corresponding to the maximum load reached in a tensile test expressed as a percentage of the original gauge length. It is also termed as uniform elongation and is recorded before initiation of necking.

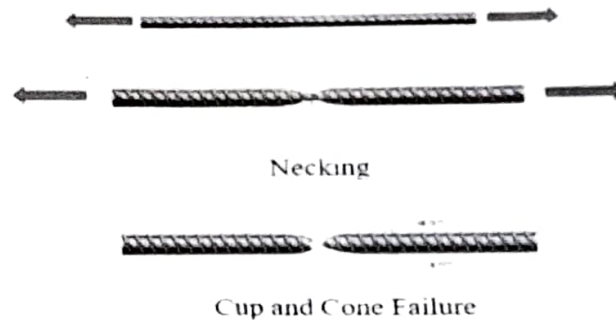


Figure 3: Cup and Cone Failure

### Test Method

#### **Determining the effective cross-sectional area**

1. Measure the total length of the specimen using a measuring tape in meter up to 3rd decimal place.
2. Measure the weight of the specimen on a weighing balance in Kg up to 3rd decimal place.
3. Calculate the cross-sectional area ( $S_0$ ) of the specimen.  $S_0 = w / 7.850L$ ,  $w$  = weight(g) and  $L$  = total length(mm)

#### **Choosing gauge length, $L_0$**

1. Determine the original gauge length from  $L_0 = k \sqrt{S_0}$ , the specimen have a direct relationship between the original gauge length and the original cross-sectional area ( $S_0$ ), where  $k$  is a coefficient of proportionality. The internationally adopted value for  $k$  is 5.65.  
 $L_0$  may round to the nearest multiple of 5 mm, provided that the difference between the calculated and chosen gauge length is less than 10 % of  $L_0$ .

### Procedure:

1. Measure the diameter of the specimen at least three different places and determine the mean value. Also measure its gauge length
2. Fix the specimen between the upper and middle cross head jaws of the machine
3. Close the release valve on the machine and turn on the machine
4. Now apply load on the specimen by turning slowly the load valve in anticlockwise direction until the specimen fails.
5. Note down the load and its corresponding length. The observations are taken until the specimen breaks.
6. Measure the final length and diameter of specimen at the fracture by joining the two broken halves of the specimen together
7. Now calculate the desired values and plot a graph between stress vs. strain.

**Observation Table: (1)**

| Sl. No. | Initial diameter 'D1' (mm) | Final diameter 'D2' (mm) | Initial length 'L1' (mm) | Final length 'L2' (mm) | Initial cross section area 'A1' (mm <sup>2</sup> ) | Area at fracture 'A2' (mm <sup>2</sup> ) |
|---------|----------------------------|--------------------------|--------------------------|------------------------|----------------------------------------------------|------------------------------------------|
|         |                            |                          |                          |                        |                                                    |                                          |

**Observation Table: (2)**

| Sl. No. | Load 'P' (kN) | Displacement 'δl' (mm) | Stress 'σ' (kN/mm <sup>2</sup> ) | Strain 'e' | Modulus of elasticity 'E' (kN/mm <sup>2</sup> ) |
|---------|---------------|------------------------|----------------------------------|------------|-------------------------------------------------|
|         |               |                        |                                  |            |                                                 |

**Calculations:**

1. Stress

$$= \frac{P}{A_1} \text{ kN/mm}^2$$

2. Strain

$$= \frac{\delta l}{L_1}$$

3. Modulus of Elasticity

$$= \frac{\sigma}{e_t} \text{ kN/mm}^2$$

4. Ultimate tensile strength

$$= \frac{\text{Ultimate Load}}{A_1} \text{ kN/mm}^2$$

5. Percentage elongation

$$= \frac{L_2 - L_1}{L_1} \times 100\%$$

6. Percentage reduction in area

$$= \frac{A_1 - A_2}{A_1} \times 100\%$$

**Conclusion:**